

PROJECT DOCUMENTATION

LifeLink — Emergency Resource Coordination Platform

Real-Time Multi-Resource Triage, Geospatial Matching Engine & Volunteer Privacy Architecture

Project Domain: Critical Healthcare Dispatch & Emergency Informatics

Architecture: FastAPI (Python 3.10) + React 19 / Vite + WebSockets + Leaflet GIS

Geographic Deployment: Bhubaneswar Emergency Medical Region (AIIMS, Capital Hospital, Red Cross)

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Core Viva Proposition:

"Our problem is not simply finding a blood donor. The problem is coordinating the entire emergency response quickly."

During critical trauma, fragmented communication across WhatsApp, phone calls, and manual searches causes lethal delays. LifeLink solves this through automated multi-factor resource ranking, verified inventory synchronization, and sub-second WebSocket dispatch.

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1. Executive Summary & Problem Formulation

In emergency medical emergencies—such as road traffic accidents, acute surgeries, or post-partum hemorrhages—access to time-sensitive compatible blood components (Packed Red Blood Cells, Platelets, Plasma, Whole Blood) and immediate support services (Ambulances, ICU beds) is fragmented across disconnected hospital telephone switchboards, social media posts, and unverified contacts.

The Concrete Dilemma: Rahul Sharma (Age 28) suffers severe trauma in Bhubaneswar and requires **2 Units of O+ Packed RBC** at AIIMS Hospital within **2 hours**. Traditionally, the family calls relatives, sends broadcast messages into WhatsApp groups, calls random blood banks, and searches Google Maps while vital minutes expire.

The LifeLink Solution: LifeLink acts as a unified command-and-control emergency exchange. A single request triggers an instant spatial query that checks nearby verified blood banks (e.g. AIIMS Blood Bank, Capital Hospital, Red Cross) and nearby registered voluntary donors, computes a normalized **Match Score**, and initiates a prioritized real-time dispatch cascade via WebSockets.

2. Technical Architecture & System Design

The system follows an asynchronous, decoupled client-server architecture tailored for sub-second emergency response times:

Layer	Technology	Role & Key Responsibilities
Client (Frontend)	React 19, Vite 8, Tailwind v4	Interactive responsive UI, dual triage actions, role portals, toast alert listener
Map Radar	Leaflet & React-Leaflet GIS	Real-time OpenStreetMap canvas plotting color-coded hospital, blood bank, ambulance, donor markers
Backend REST API	FastAPI (Python 3.10)	High-performance async request routing, Pydantic validation, JWT token creation
Real-Time Dispatch	Native WebSockets Hub	Bi-directional event bus pushing instant emergency notifications and status shifts
Matching Engine	Python Mathematical Core	Multi-factor scoring: Haversine distance, stock availability, urgency weighting
Data Persistence	SQLAlchemy 2.0 / SQLite / Postgres	ACID transactional store for multi-user requests, donor profiles, inventories, audit logs

3. Mathematical Matching & Ranking Engine

LifeLink distinguishes itself from generic CRUD systems through its quantitative ranking algorithm. When an emergency request is lodged, candidate resources receive a normalized score $S \in [0, 99]\%$:

Normalized Score Formula:

$$S = S_{\text{avail}} (35 \text{ pts}) + S_{\text{dist}} (30 \text{ pts}) + S_{\text{type}} (15 \text{ pts}) + S_{\text{urgency}} (10 \text{ pts}) + S_{\text{compat}} (10 \text{ pts})$$

1. **Distance Decay Score:** $S_{\text{dist}} = 30 * (1 - (d / 25.0) * 0.75)$ for $d \leq 25 \text{ km}$

where distance d is computed using the Great-Circle Haversine formula:

$$d = 2R * \arcsin(\sqrt{ \sin^2(\Delta\phi/2) + \cos(\phi_{\text{sub1}})\cos(\phi_{\text{sub2}})\sin^2(\Delta\lambda/2) })$$

2. **Stock Availability:** Full 35 pts if verified stock $\geq$ needed units; proportional points if partial stock.

3. **Institutional Reliability:** Blood Bank = 15 pts, Hospital = 14 pts, Verified Voluntary Donor = 12 pts.

4. **Urgency Weight:** Critical emergency requests (<2 hrs) grant maximum priority to proximate units.

5. **ABO/Rh Blood Compatibility:** Exact blood group match receives 10 pts; universally compatible units receive 7 pts.

Medical Governance Protocol: LifeLink strictly operates as a coordination and discovery platform. As highlighted in national transfusion guidelines, clinical blood compatibility verification, cross-matching, and final component release are exclusively performed by licensed medical officers at the receiving blood centre.

4. Real-time Workflow Pipeline & State Machine

An emergency lifecycle progresses through five deterministic workflow states tracked concurrently on both requester and coordinator screens:

State Code	Stage Name	Trigger Action	WebSocket Broadcast Event
MATCHING	1. Matching	Emergency request submitted by attendant	EMERGENCY_CREATED (payload with request code, units, hospital)
CONTACTED	2. Blood Bank Contacted	Matching engine contacts apex centers	STATUS_CHANGED (alerts staff of pending reservation)
RESPONDED	3. Volunteer Response	Donor reviews alert & clicks [ACCEPT]	VOLUNTEER_ACCEPTED (issues ephemeral bridge token)
CONFIRMED	4. Resource Confirmed	Unit reserved / donor en-route to center	STATUS_CHANGED (locks assigned coordinator)
COMPLETED	5. Completed	Blood delivered & transfusion initiated	STATUS_CHANGED (archives request into hospital analytics)

5. Database Relational Architecture

Entity Table	Primary Key	Key Attributes & Foreign Keys	Cardinality
users	id (Int)	name, email, phone (hashed), role, is_verified	1 : 1 donor_profile, 1 : N requests
donor_profiles	id (Int)	user_id (FK), blood_group, lat, lng, preferred_radius_km, is_available	1 : 1 with users
blood_banks	id (Int)	name, license_no, category, lat, lng, contact_phone, is_verified	1 : N blood_inventory
blood_inventory	id (Int)	blood_bank_id (FK), blood_group, component, units_available, last_updated	N : 1 with blood_banks
hospitals	id (Int)	name, registration_no, address, lat, lng, emergency_contact, total_beds, icu_beds	Apex reference node
emergency_requests	id (Int)	request_code (LL-xxxx), patient_name, blood_group, units_needed, status	1 : N resource_matches
resource_matches	id (Int)	request_id (FK), resource_type, resource_id, score, distance_km, status	N : 1 with emergency_requests
ambulances	id (Int)	vehicle_number, provider_name, ambulance_type (BLS/ALS), driver_name, is_available	Independent fleet node
pharmacies	id (Int)	name, address, lat, lng, is_24x7, inventory_notes, last_verified_minutes_ago	Independent supplier node

6. Platform Modules & Frontend Portals

LifeLink provides specialized, high-fidelity views tailored to the distinct participants in an emergency:

- **Homepage & Dual Triage Dispatch:** Features high-contrast action buttons: **'I NEED HELP'** (triggers sub-2-minute emergency request submission) and **'I CAN HELP'** (donor onboarding). Displays real-time metrics and emergency feed.
- **Emergency Radar Map (Leaflet):** Interactive spatial canvas with custom HTML markers: Blood Banks (Red), Hospitals (Indigo), Ambulances (Amber), Donors (Emerald), and Active Patients (Pulsing Red SOS Beacon) with 1-click Google Maps turn-by-turn navigation.
- **Volunteer Privacy & Masked Bridge:** Donors toggle emergency availability, calibrate response radiuses (2–30 km), and accept alerts. Under zero phone leakage rules, an ephemeral bridge token (e.g. SECURE-BRIDGE-1-2) connects the parties.
- **Hospital Triage & Analytics:** Apex dashboard monitoring active requests, critical trauma counts, and resolved cases (31 received, 26 resolved, 5 pending).
- **e-RaktKosh Aligned Blood Bank Hub:** Allows certified lab technicians to manage stock by component (Packed RBC, Platelets, Plasma, Whole Blood) with real-time increment/decrement controls that synchronize across the entire city instantly.
- **Ambulance Fleet & 24x7 Pharmacy Telemetry:** Provides vehicle dispatch triggers and rare medication searches (Heparin, Albumin, Tranexamic Acid) featuring explicit 'Reported 14m ago' freshness notices.

7. Automated Testing & Verification Results

A rigorous end-to-end automated test suite (backend/test_pipeline.py) was constructed to validate the entire coordination lifecycle:

Test #	Target Component / Verification Objective	Observed Outcome	Status
T1	Backend Health Check (/api/health)	HTTP 200 OK — service online	PASSED
T2	Geospatial Markers (/api/resources/map)	17 verified medical markers returned	PASSED
T3	Emergency Creation & Engine Execution	Request generated; AIIMS scored 99.0%	PASSED
T4	Workflow Pipeline Advancement	Transitioned to CONTACTED state	PASSED
T5	Volunteer Response & Privacy Bridge	Accepted; SECURE-BRIDGE token issued	PASSED
T6	Autonomous State Shift Validation	Status auto-promoted to RESPONDED	PASSED
T7	Hospital Triage Metrics Counter	Active queue accurately updated	PASSED

8. VS Code Execution Guide & 5-Minute Viva Demo

Step-by-Step Launch Instructions:

1. Open folder in VS Code: File -> Open Folder -> lifelink
2. Press **Ctrl + Shift + B** to execute the pre-configured compound task '*Start LifeLink (Full Stack)*'.
3. Alternatively, use split terminals:
 - Terminal 1 (Backend): `cd backend && python run.py` (API: `http://127.0.0.1:8000`)
 - Terminal 2 (Frontend): `cd frontend && npm run dev` (UI: `http://localhost:3000`)

Examiner 5-Minute Demonstration Script:

- *Minute 1:* Present the homepage and discuss the fragmentation of emergency care.
- *Minute 2:* Submit a Critical Request for 2 units of O+ Packed RBC at AIIMS Hospital.
- *Minute 3:* Demonstrate the Matching Engine results ranking AIIMS (99.0%) and Capital Hospital (94.7%).
- *Minute 4:* Open the Volunteer Portal, display the incoming emergency alert, and click [ACCEPT DISPATCH].
- *Minute 5:* Show how the requester's workflow stepper automatically moves to 'Volunteer Response' live via WebSockets.